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1 Project Group ID: G-11

2 Title of Project: Sign Language Recognition Using Deep Learning

3 Domain: Deep Learning

4 Abstract

People who are unable to talk or hear can only communicate through sign-language. People who are physically disabled can communicate their thoughts and feelings by using sign language. In this work, a system for recognising sign language has been proposed in order to recognise the alphabets and gestures used by disabled community. With the aid of neural networks, we can recognise the signs and provide the appropriate text output, bridging the gap between physically challenged people and average people.

5 Introduction

Since the majority of people in the world do not know sign language and finding interpreters is very difficult, we have developed a real-time method using neural networks for American sign language based on hand gestures. Sign language is one of the longest living and most natural forms of language in communication.By using gesture formed by different hand shapes deaf-dumb people can communicate with us.

The syntax of American Sign Language (ASL) differs from that of English although it is a fully developed natural language with the same lexical elements as spoken languages. In ASL, communication is done through hand gestures. It is the majority language of both hearing and deaf North Americans, including those who are hard of hearing. As a result, it improves the communication between physically disabled persons and other people..

5.1 Background

People who are mentally impaired frequently lack access to regular social interaction. Additionally, it's challenging for regular individuals to understand and interact with them. These individuals are forced to use visual communication or an interpreter. It won't always be possible to use an interpreter, and visual communication is typically challenging to understand.

5.2 Purpose

In the community of the blind and mute,using signs is the main form of interaction. Making these folks feel included in society in today's technologically advanced world requires facilitating their communication. This means that an intelligent computer system needs to be created and trained. For a while now, researchers have been working on the issue, and the outcomes appear promising. Although modern voice recognition technologies are being created, the industry currently lacks any genuine commercial products for sign recognition.

6 Advantages:

1) It is simplest way of communication for mute and blind people to interact with sighted people..
2) All children can acquire communication skills by being familiar with sign language concepts and it will help them to increase their IQ level.
3) It can be used in noisy environments or when oral speech is not possible.

7 Disadvantages:

1) For actions to be understood in sign language, the hands must be fully functional. This presents a challenge for those who are hand-impaired.
2) It requires time to learn and is unable to detect mistakes.
3) In sign language, actions must be made with the hands.

These are all the pros and downsides that have been discussed.

8 Problem Statement

To design and develop a platform that communicates information to common people by recognising various gestures.

8.1 Ideal:

A system that recognizes sign language with high accuracy and gives the output in alphabets.

8.2 Reality:

There are various existing system available including mobile applications for assistance of dumb and deaf which are built using raw datasets,so they have low accuracy.

8.3 Consequences:

People who are deaf or dumb will not be able to communicate themselves clearly in front of regular people.If a high-accuracy system is not constructed,then it increases the chances of incorrect output prediction.

8.4 Proposal:

The project aim is to create a responsive website which will give an high accuracy output for a particular sign in number as well as alphabets format on the screen.

9 Objectives

1) To insert ASL Dataset.
2) To apply image processing and segmentation to the input images using CNN technology.
3) To train and test model using Deep Learning algorithms.
4) To analyse the performance and provide respective output.

10 Brief Literature Review

A) Articles from Inter-national Journal of Engineering and Technology

a) **”Review Paper on Sign Language Recognition for The Deaf and Dumb”** One of the non-verbal communication techniques used in sign language is the hand gesture. It is mostly used by deaf and dumb people to communicate with other people or among themselves when they have hearing or speech issues. Many makers around the world have created numerous sign language systems, however they are neither adaptable nor economical for end users. As a result, the software shows a system prototype that can automatically detect sign language to aid deaf and dumb individuals in communicating with each other and other people more efficiently.

b) **”Sign Language Recognition”** by Kalsh, Er Aditi, and N. S. Garewal (2019)[2]This research paper states, By keeping in mind the fact that, in typical circumstances, every human being has the same hand form with four fingers and a thumb, this study seeks to develop a real-time system for the recognition of several significant shapes made using hands. There is a camera and computer system. Limitations: An input image’s background must be free of outside objects. Additionally, a fixed distance is maintained between the camera and the image.

c) **” American Sign Language Translator”**by Authors Andra Ardiansyah,Brandon Hitoyoshia(2020)[3] The authors have found 22 relevant research papers between 2015 and 2020 that deal with enhancing automated American Sign Language translators. An analysis of the papers revealed that while the studies selected had respectable outcomes, they were not flawless because each study revealed its own distinct strengths and weaknesses.Some techniques, such as using a regular video camera to collect data and either a convolutional neural network or a support vector machine for classification, may be appropriate for developing a useful sign language translator.

11 Methodology

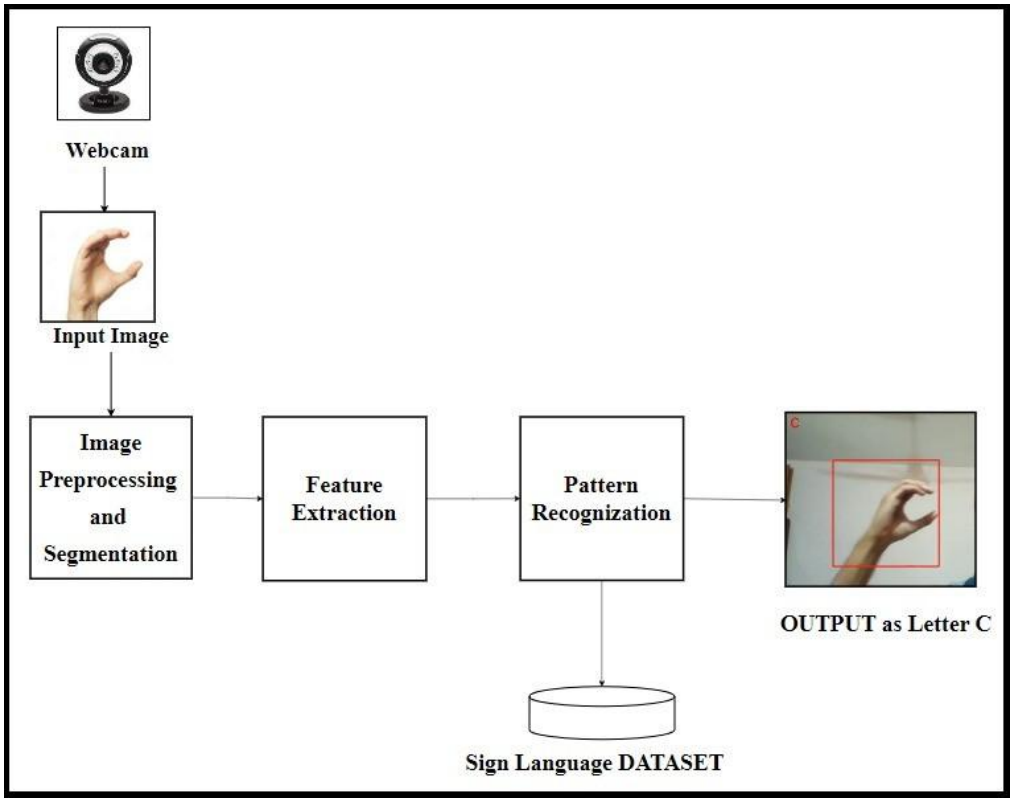

Fig.1 : Methodology of system

11.1 Image Pre-processing

We will create a image classifier which can distinguish whether a detected hand gesture is of which sign.The widely utilised machine learning algorithm CNN, which is used for image classification, will be employed to help us accomplish our goal.The CNN model receives the processed image and begins training.

CNN- A deep learning system called a convolutional neural network (CNN) may evaluate an image as input, classify its items according to significance, and tell them apart from one another. The pre - processing required for ConvNet is matched with alternative classification algorithm.The filters are added by primitive methods, but a ConvNet can learn those filters with enough practise.

11.2 Segmentation

Segmentation is used to simplify and improve the comprehension of the visual depiction. Deep learning algorithms are the foundation of contemporary picture segmentation methods. The process of image segmentation divides a picture into a number of pixel sections that are each represented by a mask or labelled image. You may process only the crucial portions of the image by segmenting it rather than processing the whole thing.

Another typical strategy is to look for commonalities across different areas of photos. This method is followed by some approaches, such as region expanding, clustering, and thresholding.

11.3 Feature Extraction

After pre-processing has produced the required level of segmentation, features are extracted from the segments using a variety of feature extraction approaches. The application of categorization and post-processing strategies comes next. It is essential to concentrate on the feature extraction phase since it can be thought to influence how well the recognition system performs. The feature selection of a feature extraction approach is the single most important factor in obtaining strong recognition performance. ”Extracting from the raw data information that is most relevant for classification purposes, while reducing the within class pattern variability and enhancing the between class pattern variability,” is how feature extraction is defined. As a result, extreme care must be used while choosing a feature extraction technique that is appropriate for the input being applied. It is crucial to evaluate the numerous feature extraction approaches that are accessible in a specific domain that covers a wide range of potential scenarios while taking all of these elements into account.

11.4 Dataset:

For this project, we used an already-existing dataset in American Sign Language (ASL) from the Nikolas-created kaggle website. The dataset consists of cropped RGB photos of ASL (American Sign Language)handshapes that correspond to the 26 letters of the English alphabet and space, nothing, and delete for further scope. The dataset has a total of 87,000 images, with 3000 images for each alphabet.

11.5 Pattern Recognition:

Pattern recognition is the process of finding patterns using a machine learning model. Pattern recognition is the categorization of data based on existing knowledge, statistical data derived from patterns and their representation, or both. After picture extraction, the provided input image is compared to the current ASL Dataset. We print a letter whenever the sign of a letter recognises a particular gesture and no other letter is close to it by a threshold. In that it uses CNN and other necessary algorithms like the k-nearest neighbour algorithm, etc., this technique is similar to deep learning.

12 Requirements

12.1 Hardware requirements:

1) Webcam:Excellent HD camera quality
2) Ram: Minimum 8GB or higher
3) Graphical Processing Unit: 4GB dedicated
4) Processor:at least an Intel Pentium 5

12.2 Software requirements:

1) Operating System: Windows/Mac/Linux
2) Source Developement Kit: OpenCV,TensorFlow,Keras,Numpy,PyTorch

13 Implementation:

Steps to be Followed:

13.1 Production of the preprocessed image.

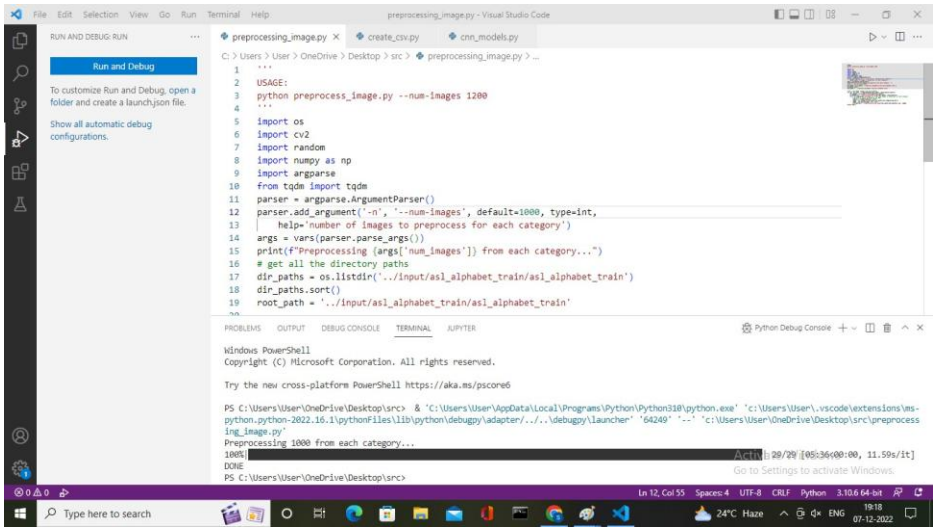

Fig.1 Preprocessed images

We will write the code to preprocess a portion of the 87,000 total images in this section. Each class has 3000 images total. However, from each class only 900 images will be used. Therefore, there will be 29000 images in all.

13.2 Setting up the data-csv file.

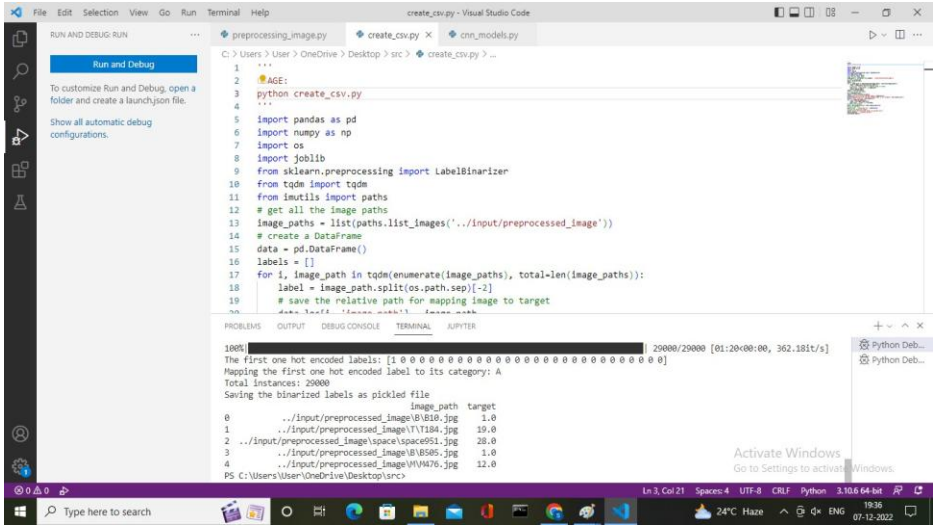

Fig.2 data-csv file generation

The data-csv file it has two columns. The first is the image path column, which will include each and every image path. The target column, which will be in the second column, will contain a number between 0 and 28 that designates the image's class.

13.3 Adding the cnn models.py file with our neural network design.

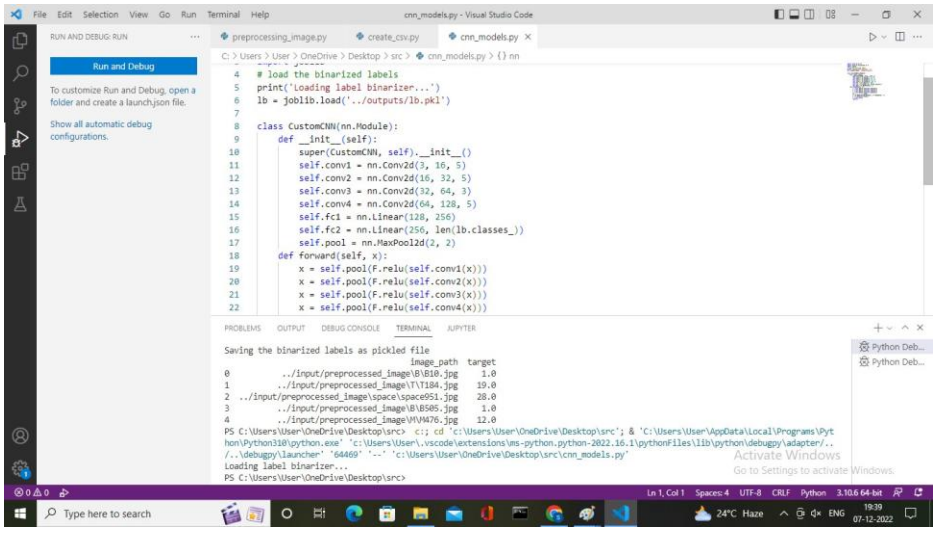

Fig.3 cnn models Processing

In the file cnn models.py, we will write the code. It is quite helpful to keep the designs of deep neural networks in a distinct Python script. Multiple class modules can be written, and many neural network designs are possible. You can simply import any architecture you want to employ into the training file.

13.4 Inserting the training code into the train.py file.

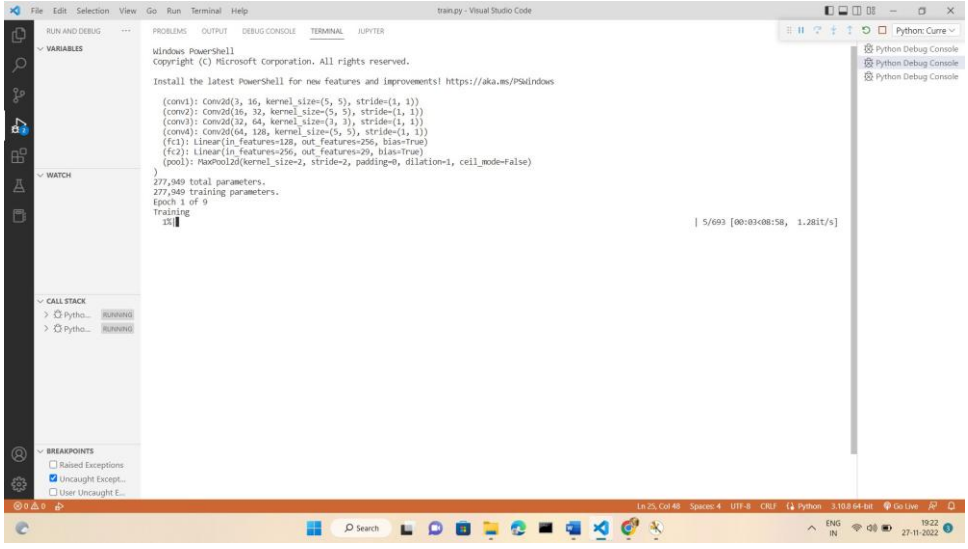

fig 4.Training Epoch

The neural network model will then be trained on the preprocessed images using training code, which will be written next. About 15 per of the information is being used for verification.It is due to the size of our dataset, which includes a sizable amount of photos for validation in 15 percent of the data.

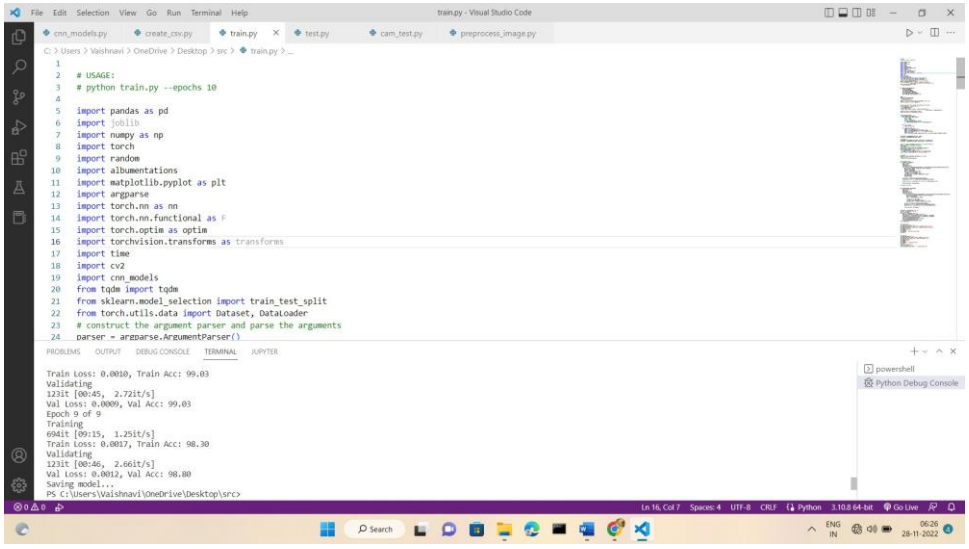

fig 5.Saved model

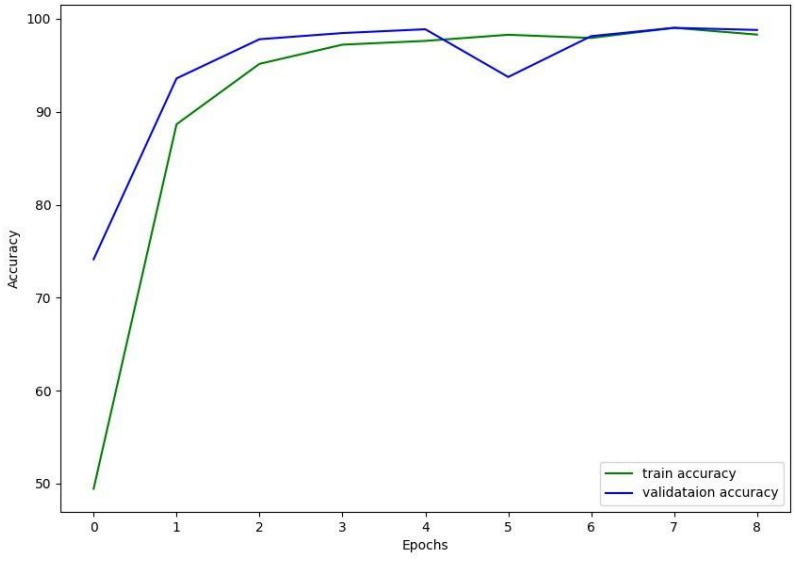

fig 6.Accuracy graph

Up to the end of training, the accuracy rates for training and validation were both above 90 percent . Given how simple our deep learning model is, we eventually achieve better than 98 percent accuracy, which is fantastic.

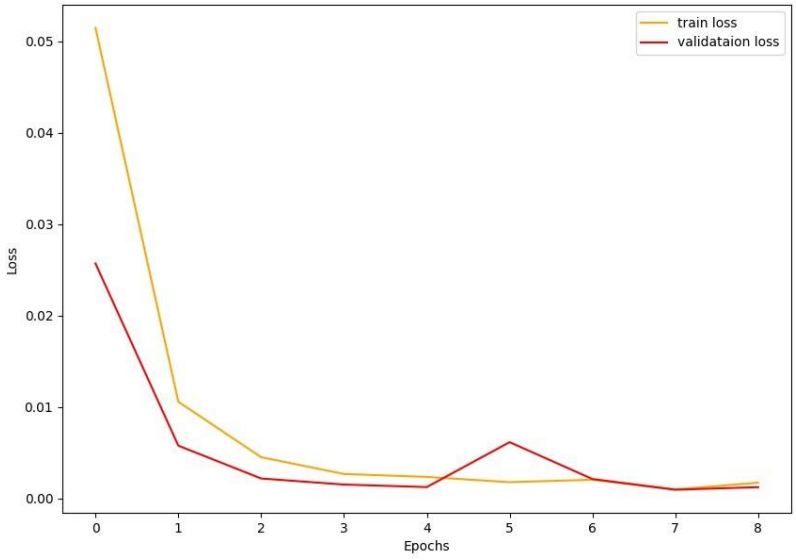

fig 7.loss value graph

Additionally, the loss values look excellent. The training loss and the validation loss are closely related. By the time the training is over, we will have lost almost nothing.

13.5 Writing the test code.

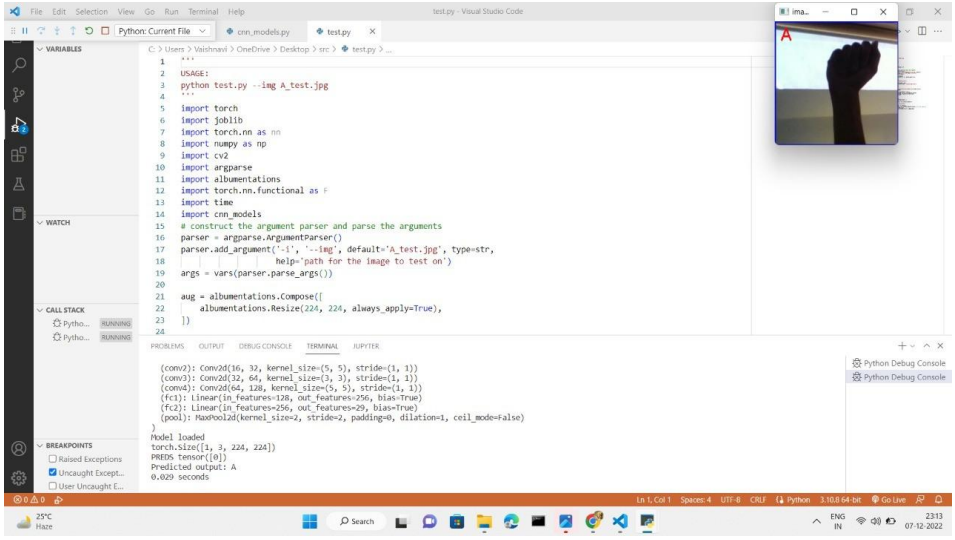

Fig.8 Testing process

In this section, we'll advance the code that will let our deep neural network examine the test images. Within the input folder are them. We obtain the photos via the code input that we will load. The image will then undergo the agumentation, be converted into a torch tensor, and sent to the CPU for testing.

13.6 Then, for the live camera stream, we will create the code for identifying the sign language letters within the cam test.python file.

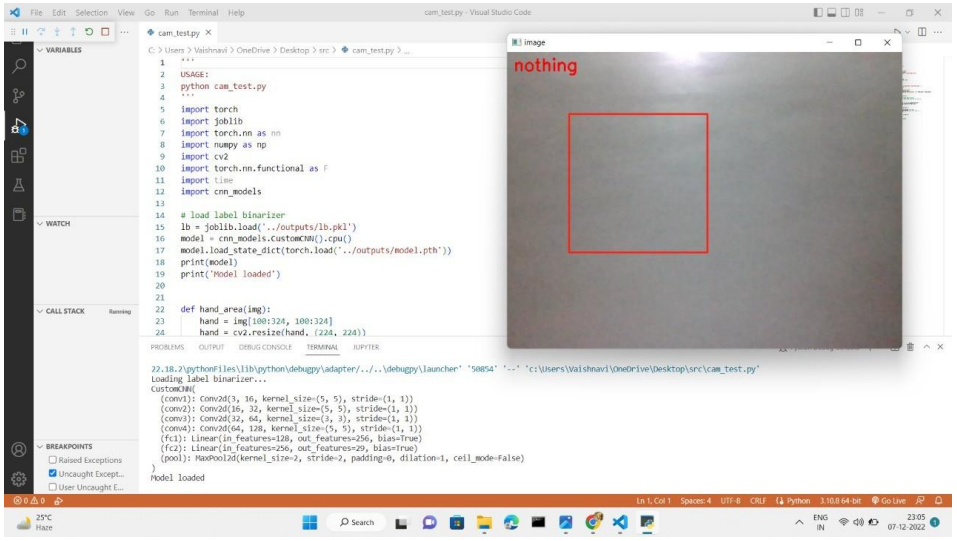

Fig.9 Opening Webcam

Connecting our Python code and reading camera frames is made really simple by OpenCV. We must have a webcam attached to our computer in order to complete this step.

14 Results:

Result is shown through following :

14.1 Result for Letter A:

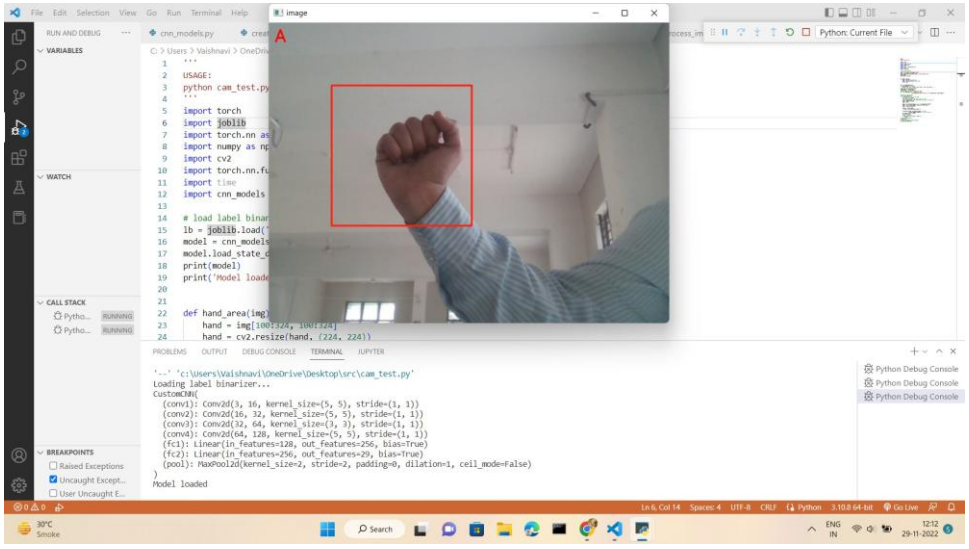

Fig.Output as A

14.2 Result for Letter B:

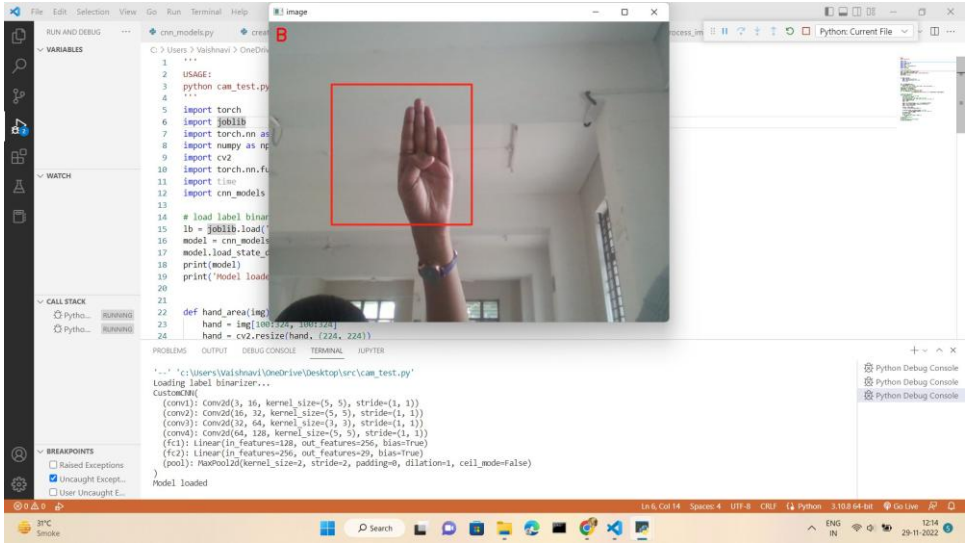

Fig.Output as B

15 Conclusion:

When we consider all of the potential combinations of motions that a mechanism of this type must interpret and convert, sign language recognition becomes difficult challenge. As said that, it may be wise to divide this challenge into smaller challenges, with the approach provided here offering a potential solution for each of them. Convolutional neural networks and just cameras were used to build the system, which performs too well for a first-person sign -language translation system. Neural networks with convolutions can be employed as a classification system in order to identify sign-language, we conclude. Install a dataset first into the system. However, preprocessing was done on the ASL dataset's photos initially. The data csv file for the image path is then created. CNN's modules Processing occurs before the training component. Only 15 percent of the dataset is used for training, and that portion is processed, resulting in two graphs—a loss graph and an accuracy graph—both of which display excellent performance. Here, 98 percent accuracy is the highest we can get. After the training is finished, the dataset is tested, and then the OpenCv idea is used to turn on the webcam. In the end, we receive the results as alphabets.

16 Future Scope:

The suggested sign language recognition system can be expanded to recognise facial expressions and gestures in addition to sign language letters. Instead of using letter labels, statements will be more widely accepted as a highly accurate interpretation of the language. The range of several sign languages can be expanded. The letter can be detected with greater accuracy by adding more training data. The idea can be expanded further to speak the signs.

17 References

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2) Reda MM, Mohammed NG, Azeem RAR, Seoud A (2018) SVBicomm signs voice bidirectional communication systems for normal, deaf and dumb and blind people based on machine learning. In: 1st international conference on computer application and information security (ICCAIS). IEEE, Riyadh
3) Kalsh, Er Aditi, and N. S. Garewal. ”Sign language recognition system.” International journal of computational engineering research 3, no. 6 (2017): 15-21
4) M. I. Saleem, P. Otero, S. Noor and R. Aftab, ”Full Duplex Smart System for Deaf and Dumb and Normal People,” 2020 Global Conference on Wireless and Optical Technologies (GCWOT), 2020, pp. 1-7, doi: 10.1109/ GCWOT49901. 2020.9391593.

18 Project Plan :

Table 1 : Schedule

Month	Project work/Plan
August 2022	Project Proposal and Topic Selection
Septmber 2022	Synopsis and requirement analysis
October 2022	Planning and Designing
November 2022	Coding and Application Building
December 2022	Project Report

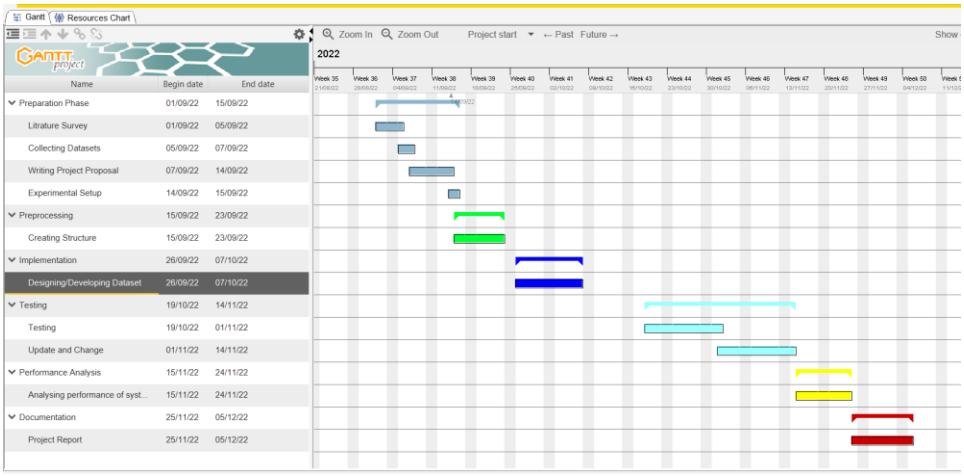

Fig.Project Scheduling

19 Group Members :

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Mr.S.I.Tamboli
Project Coordinator

Mr. Suhel Sayyad,HOD(CSE)